It's Like a Dark Patterns Paradise: New Findings from Brazilian Mobile Apps in Light of International Research

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Abstract. Research on Dark or Deceptive Patterns (DPs) primarily focuses on WEIRD (Western, Educated, Industrialized, Rich, Democratic) contexts, leaving the understanding of DPs in the Global South largely unexamined. This study analyzes DPs in 100 trending Android applications (apps) in Brazil. The results showed that 96% of the apps have at least one DP, averaging 7.51 DPs per app, and 46% of them contain at least 7 DPs. We identified six previously unreported DP cases in the Brazilian context, demonstrating novel applications of known DPs. We compared our results with previous studies conducted in Japanese and American contexts, examining common subclasses and approaches to addressing the identified differences in culture, wealth, and education.

Keywords: Dark Patterns · Deceptive Patterns · Deceptive design · Manipulative design · Mobile Applications · Brazil · Global South · Ethical design · Human-Computer Interaction

1 Introduction

Dark Patterns (DPs) are user interface design choices that lead users to make decisions they might not have made otherwise, often for the benefit of an online service [9]. DPs often result in user behaviors that are against their best interests, such as impulsive buying and installing applications (apps) that they do not need [2]. The nature and impacts of DPs have been drawing attention from the Human-Computer Interaction (HCI) community [7]. Although numerous studies on DPs have focused on categorizing [13,8] and identifying the impacts [18,14] of these patterns, those occurred primarily within Western and Anglocentric societies [17], so it is crucial to understand how different countries recognize and experience DPs in their interactions with digital interfaces.

For instance, when replicating Di Geronimo's procedure, Hidaka et al. [20] discovered a new class of DP called "Linguistic Dead Ends", which discourages or prevents users from understanding the interface and/or content of an app

by manipulating language at a fundamental level. Hidaka's findings reveal that DPs were less prevalent in the Japanese context, with an average of 3.9 compared to 7.4 found on Di Geronimo's report. Consequently, our understanding of DP prevalence and manifestation, often based on taxonomies developed in these contexts, may not fully capture the situation in the Global South. As such, the topics studied in DPs may differ in non-WEIRD countries.

Further, determining whether the Brazilian context is WEIRD or non-WEIRD is difficult, due to its complex social indicators [11]. The complexity of this situation creates new challenges for the HCI community. Thus, it presents a critical context for investigation of DPs impact, representing a large digital market within the Global South with specific characteristics in questions such as inequality in education [5], lack of digital literacy [21] and uneven spread of Information and Communication Technologies (ICTs) [10], because of that we hypothesize that DPs in Brazilian context have particularities.

Inspired by the culture-specific DPs found in Japan, which are linked to linguistic characteristics [20], this work explores context-specific aspects in Brazil regarding DPs. We aim to present baseline data on the Brazilian prevalence and types of DPs, establishing a point of comparison with results from research studies in the United States [3] and Japan [20]. Specifically, we investigate (i) the frequency and types of DPs in the most widely used Brazilian apps, categorized according to established classifications; (ii) how they contrast with previous research conducted in other geographical settings.

To this end, we adapted the qualitative expert classification protocol of Di Geronimo et al. [3] and included the DP class introduced by Hidaka et al. [20]. We analyzed the top 100 trending [1] Android apps in Brazil across 9 categories, classifying identified DPs based on a taxonomy derived from Gray et al. [9], Di Geronimo et al. [3], Hidaka et al. [20] and additions we made to it: 6 new DP cases found in the Brazilian context. Our work contributions include: (i) Identifying a baseline DP prevalence in popular Brazilian mobile apps (96% affected, averaging 7.51 DPs/app).; (ii) Identifying higher rates of Nagging and Forced Action in Brazil compared to results from United States [3] and Japan [20]; (iii) Presenting and discussing the prevalent DPs in Brazil and the notable presence (45% of apps) of Untranslation, offering context-specific insights for future Global South research on DPs.

2 Methodology

We replicate the methodology of Di Geronimo et al. [3] with a focus on the Brazilian context to identify differences and similarities in the presence and frequency of DPs in the Brazilian apps. However, we adjusted the procedure of the original study to address limitations and incorporate emerging findings from our own study. The resultant database is available on Zenodo [16].

[1] The term "trending" refers to apps ranked highly in the Google Play Store based on ratings, reviews, and downloads.

2.1 Adaptations from the Original Protocol to Our Context

With the following adaptations, our work provides an initial classification of DPs in trending Brazilian apps and a comparison with previous studies from other countries [3,20].

Corpus Generation. We used nine categories by adding "Tools", compared to the eight in Di Geronimo's study. We also reduced our sample to 100 apps, fewer than Di Geronimo's 240.

Recording and Classification. We added a test phase to determine our recording method and find necessary protocol changes. For this, recording and classification procedures of Di Geronimo's study [3] were performed for two pairs of researchers in 9 different apps. One pair used their own Android smartphones, and the other pair used an emulator. We selected the emulator as the official method to keep consistency between devices and budget limitations. Testing shows that DPs were detectable in emulator recordings as well as in smartphone recordings. Based on this test, we added the "checking the terms of service and privacy policy" task. This phase helps to consolidate consistency in the analysis between our two researcher pairs (utilizing a rotation method, different from Di Geronimo's single pair) and leads to identifying potential new findings. Apps used in the test were re-recorded by a different researcher (to avoid bias) and used in the final analysis.

New Usage Contexts. We added the macro class Linguistic Dead-Ends and its subclass Untranslation, discovered by Hidaka et al. [20] to our study because of their presence in the test phase. Furthermore, we identified new findings of specific contexts of DP not reported by Di Geronimo, resulting in six new cases of DP described in the results section.

2.2 Corpus Generation and Recording of Usage

To generate a corpus for the study, we first examined the top 100 trending general apps in the Brazilian Google Play Store [6] on January 19th, 2025. We searched for the uses of the app, suspicious reviews, and the most frequent categories in the 100 most popular apps, then we tried to fit those apps in the categories reported by Di Geronimo (Communication, Entertainment, Family, Music & Audio, News & Magazines, Photography, Shopping, and Social).

In this initial overview, we noted the presence of the app categories Tools and Finance, which aren't within the scope of Di Geronimo [3]. We added them to our scope since each represents 16% (16 apps) of the top 100 trending apps.

During the recording phase, we discovered that the Finance category doesn't work on emulators, so we removed it from the future steps. We used Sensor Tower for the final selection [19] on January 22nd, 2025, following four criteria: (i) It must be available to Android devices; (ii) It must be free to download; (iii) It must be trending in Brazil; (iv) It must be trending in its category.

We did not include apps that were: (i) Not available for download; (ii) Not available in Brazil; (iii) Included in another category; (iv) "Launcher" Apps; (v)

Required a subscription to access its main features; (vi) Tied to the citizen's ID and used to access governmental services.

We approved 12 trending apps in each category for the analysis phase. Some apps malfunctioned in the emulator, resulting in the analysis approval of 100 apps divided into the following structure: 12 in Family; 12 in Music & Audio; 12 in News & Magazines; 12 in Photography; 12 in Social; 11 in Shopping; 11 in Tools; 9 in Communication; 9 in Entertainment.

We used BlueStacks 5 [1] to emulate the Samsung Galaxy S22 Ultra and run the apps. To capture the screen, we used a native tool on desktops powered by Windows. Each researcher created a new Gmail account. We also purchased a SIM card and plugged it into a smartphone to validate its use in apps that required a phone number. To ensure consistency in the usage, we recorded the first 10 minutes of the selected apps following eight tasks (when available): (i) creating an account and log out; (ii) closing and reopening the app; (iii) visiting the market page; (iv) going to the setting page; (v) checking the terms of service and privacy policy; (vi) continuing shopping until checkout; (vii) trying to select product names in e-shopping; (viii) using the app for its intended use. We did not purchase any products or services in the apps.

2.3 DPs Classification

We organized ourselves into two pairs. While one pair was recording the usage, the other was analyzing the videos. Once we had already analyzed the first half of the apps, we switched roles. During the classification process, we first analyzed the videos individually. Then, we discussed them together to mitigate DP Blindness. The agreement between the pairs was good according to Weighted Cohen's Kappa ($\kappa = 0.64$) which indicates substantial agreement. We included a third researcher to break ties when the pair did not reach an agreement.

Di Geronimo's expansion [3] of Gray's taxonomy [9], together with Hidaka's later findings related to linguistic DPs [20], based our classification process. Therefore, the taxonomy used in this study divides the DPs into six macro classes: (i) Nagging; (ii) Obstruction including three subclasses (Intermediate Currency, Price Comparison Prevention, and Roach Motel); (iii) Sneaking including four subclasses (Bait and Switch, Hidden Information, Preselection, and Sneak into Basket); (iv) Interface Interferences including five subclasses (Disguised Ad, Aesthetic Manipulation, Toying with Emotions, False Hierarchy, and Trick Question); (v) Forced Action including two subclasses (Privacy Zuchering and Disguised Ad); (vi) Linguistic Dead-End including the subclass Untranslation. Besides the macro class Linguistic Dead-Ends defined by Hidaka [20], the macro classes and subclasses are defined in detail in Gray's study [9].

Alongside the types of DPs, we also followed a guide of 40 specific DP cases [16] to address the subjectivity of identifying DPs based solely on their descriptions. There were DP cases that fit more directly with the macro class itself. For this reason, Nagging, Obstruction, and Forced Action are macro classes that also function as unique subclasses within themselves. Additionally, to en-

sure that using the app would not interfere with the appearance of DPs, we did not count re-occurrences of DPs that appeared early on the app.

3 Results

Among the 100 studied apps, 96% of them contained at least one DP, and presented a total of 751 DPs across all corpus, with an average of 7.51 DPs per app (std.dev.:4,7), very similar to what was found by Di Geronimo [3] (avg.: 7.4, std.dev: 5). Additionally, 13% (N=13) of the apps contained between 0 and 2 DPs, 41% (N=41) contained between 3 and 6 DPs, and 46% (N=46) of them contained at least 7 DPs.

Between the six macro classes, apps contained an average of 3.7 macro classes per app (std.dev.: 1.5). The most frequent was Nagging DPs appearing in 78% (N=78) of the apps, followed by Interface Interference with 76% (N=76) and Forced Action with 62% (N=62), as shown in Fig. 1.

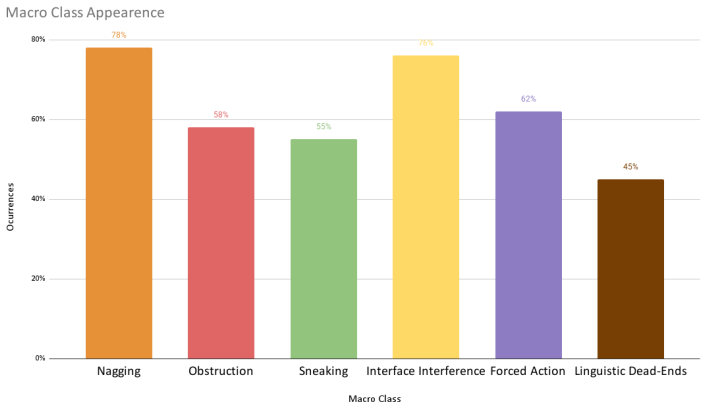


Fig. 1. Percentage of Appearance of Macro class in the Corpus

The subclasses appear in an average of 4.4 per app, with the most common being Nagging with 78% (N=78), Forced Action with 54% (N=54), and Obstruction with 50% (N=50), Fig. 2. Since we added two DP cases to the last, it became a functional subclass. Besides, the DPs with the most total appearances were Nagging (N=201), Forced Action (N=115), and Disguised Ad (N=75), also considering the multiple times it could appear.

The app with the most cases was "*Notícias Globais*" with 25 DPs between 7 different subclasses. Only 4 apps had no cases, 2 of them were from the "News and Magazine" category.

As one app category may contain more DPs than another, we ran a Kruskal-Wallis test ($\chi^2 = 15.4$, p = 0.051), as our data failed the normality assumption.

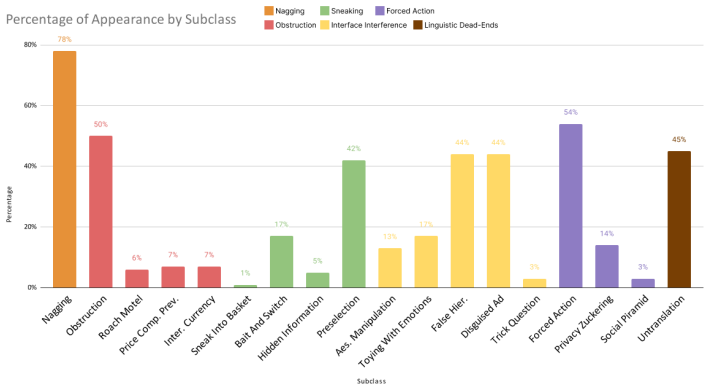


Fig. 2. Percentage of Appearance of Subclass in the Corpus

We did not find any statistically significant difference in any app category. Di Geronimo and Hidaka also failed the test and did not find any correlations between the number of DPs and the app categories. This test supports the hypothesis of the pervasive nature of DPs. A Kruskal-Wallis test ($\chi^2 = 520$, $p < .001$) revealed a significant difference between the DP subclasses. A Dwass-Steel-Critchlow-Fligner pairwise comparisons showed that the *Nagging subclass is significantly higher* than all other subclasses, except when comparing to Forced Action (W = -5.6466, p = 0.008).

Our findings also revealed six DP cases previously unreported in international research. They demonstrate novel applications of known DPs: (i) "**Terms/Policy in Another Language**" (Untranslation case) when crucial information for the user (terms of service or privacy policy) is in another language, even if the app is in the native user language, creating a barrier between the user and the information; (ii) "**Terms/Policy Out of Bounds**" (Obstruction case) redirects the user to an external page, typically a web page, to access the terms of service or privacy policy, increasing unnecessary stress and reducing the likelihood of reading; (iii) "**Ads with layers**" (Obstruction case) specifies chains of multiple time-delayed sub-ads to extend user exposure within the same content, making more than one "attention call" in the same time span; (iv) "**Task-based Rewards**" (Forced Action case) describes cases where the app presents a mission beyond the main flow, that when completed rewards the user with some aspect of the app services; (v) "**Ad-locked Rewards**" (Forced Action case) indicates when the user is forced to watch an advertisement to receive a reward like virtual coins, time accelerators, or other bonuses; (vi) "**Evoke Emotions to Take Action**" (Toying with Emotions case) specifies uses of any language or other appealing visual elements to persuade the user into an action. Those DP cases occur 158 times in our classification, accounting for approximately 21% of the 751 DPs found presenting specific occurrences of: Terms/Policy in Another Language (45 times); Terms/Policy Out of Bounds (26 times); Ads with Layers (42

times); Task-based Rewards (20 times); Ad-locked Rewards (14 times); Evoke Emotions to Take Action (11 times).

Real-world examples of each of the new DP cases are available in the study's database on Zenodo [16].

4 Discussions

The discovery of six new cases of dark patterns, which constitute approximately 21% of the reported DPs, provides a more specific and consistent classification of these patterns. This new classification aligns the previous list of DP cases more closely with the Brazilian context. Additionally, it suggests that apps in Brazil (whether local or international) are employing innovative strategies to discourage usage and prompt users to take action, beyond merely displaying ads or pop-ups. These strategies often involve creating new tasks that engage users in activities beneficial to the developer, such as enabling notifications for app updates or using persuasive messages to encourage users to rate the app.

Our results indicate that the presence of DPs in Brazilian apps aligns closely with those in American apps, with an average score of 7.51 compared to Di Geronimo's 7.4, and a prevalence of 96% versus Di Geronimo's 95%. However, the distribution of these results in the DP subclasses differentiates these two scenarios. When analyzing the most recurring subclasses, it is interesting to note that even though Nagging is in first place in both surveys, Brazilian apps stand out in the percentage difference (34,5%).

A broader perspective on this distribution difference reveals that in Brazil, apps often use DPs that aim to manipulate users by directing them to alternative activities, while in the US, they tend to manipulate users by tricking them into making specific decisions and taking certain actions. The three most common DPs tried to obstruct or force the user to interact with the application. Therefore, there is an increase in the number of cases involving Nagging and Forced Action in the Brazilian context. Moreover, the fourth most common is Untranslation (Linguistic Dead-Ends), appearing at least once in 45% of the apps.

However, when comparing our results with Hidaka's findings [20], we found an increase in presence per app on 10 out of the 16 subclasses considered in the analyses. This suggests that the Japanese DP situation appears to be handled more strictly compared to the Brazilian one. This could be due to cultural, legal, and market differences. Further, the subclasses with the most increase were Aesthetic Manipulation (171,40%), Unstranslation (100%) and Trick Question (100%). The most decrease being Hidden Information (-116,60%), Roach Motal (-100%), and Intermediate Currency (-60%). Furthermore, the substantial increase in the Unstranslation subclass in the Brazilian context draws attention to a misuse of language that intentionally obscures information by utilizing a foreign language, even when the rest of the app is in the user's native language. Those differences, particularly the Untranslation, should be addressed in future works about DPs in Brazil, since the population has a score considered low in proficiency tests such as the EF English Proficiency Index [4].

The comparison of our findings with those of Di Geronimo and Hidaka reveals differences in the expression of DPs in each context (e.g., the DP occurrence surplus in 10 out of 16 subclasses, compared with Hidaka's findings). This suggests that various socioeconomic, educational, and sociocultural factors may influence how, where, and when DPs are manifested. Future research conducted in various contexts should expand this discussion and provide new insights into how cultural differences influence the presence of DPs.

5 Limitations

Our scope contains trending free apps on the Brazilian Google Play Store. We recorded 10 minutes of the first use of each app; any deviation from this predetermined scope may generate new and different results. Moreover, some interactions—and consequently DPs—might be limited or vary in an extended usage scenario. Despite the sample reduction, we found similar results, indicating that the impact was not substantial.

The intention behind using the emulator was to replicate the experience of a real smartphone. However, a specific subset of DPs may not appear in emulated applications. In certain instances, specific apps were inaccessible because they flagged the emulator as suspicious for usage.

6 Conclusion

After analyzing 100 trending mobile apps in Brazil, we found an average of 7.51 DPs/app and six new cases of use of known dark patterns in the Brazilian context. The results showed key differences compared to contexts in the United States [3] and Japan [20] (e.g., average DPs per app, common DP subclass). Our findings enhance the understanding of DPs and their consequences in societies beyond the WEIRD classification, promoting a more comprehensive view of the topic, especially within Brazilian culture. Future work should explore the presence of DPs in Finance apps and expand research by using a larger corpus that considers more representatives of the Global South and their unique aspects as non-WEIRD societies. We encourage the HCI community to broaden its perspective and actively research how DPs affect society's economy, population well-being, and culture beyond the WEIRD spectrum.

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